

GOVINDA CHAUHAN

COMPUTER SCIENCE STUDENT | LEARNING ML SYSTEMS, BUILDING SOFTWARE & AGENT TOOLS

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ABOUT

I am a Computer Science and Data Analytics student at IIT Patna. I learn by building: I moved from full-stack software into autonomous-agent systems and then into ML inference research. I also turn what I learn into practical explanations, exercises, and tools for other students.

EDUCATION

Indian Institute of Technology Patna BS, Computer Science & Data Analytics · Expected 2026

WHAT I AM LEARNING THROUGH RESEARCH

Working-Set Routing for Memory-Constrained MoE Inference WORKING PAPER · 2× TESLA T4 · 2026

- I tested expert retention, 64-to-32 fusion, router recovery, and exact paging on a measured 6.78B-parameter OLMoE parent. My strongest static run reduced allocated VRAM by 53.7%, reached 1.08× suite throughput, and retained 80.7% of an exploratory within-run quality score.
 - I learned that removing weights is not the same as preserving their computation: static removal damaged behavior, while exact paging kept the output exact but became transfer-bound. This now guides my confidence-aware ActiveSet runtime design.
- [Research note](#) · [Paper PDF](#)

AETHER PHYSICS-FROM-VIDEO RESEARCH ARCHITECTURE

- I explored how ordinary video could become a simulation-ready model through object tracking, kinematic discovery, constraint inference, and marker-free physics extraction.

SYSTEMS I BUILT OR DESIGNED

KERNEL FINANCIAL RUNTIME · LIVE

- I built a debugger for autonomous purchasing: replay and fork trajectories, test assertions, and turn verified behavior into scoped Prava permissions.

Infinity RUST-FIRST SESSION HYPERVISOR · SYSTEMS DESIGN

- I designed transactional handoffs across coding-agent harnesses while preserving workspace state, artifacts, journals, and execution continuity.

Delegators SWE Agentic API ROUTING · STREAMING · PAYMENTS · LIVE

- I built a pay-per-session coding-agent platform with compatible API access, provider routing, streaming, quotas, admin controls, payments, artifact workflows, and deployment checks.

LINKED PROJECT INDEX

Doorway — local terminal-agent cockpit	Remote-os — browser-based desktop shell
VideoUnit — generated-video evaluation	Kairo — Rust/MCP safety concept
Plans — Delegators access plans	Workbench — agent artifact workspace
CEO Hunt — founder discovery tool	Flow Pitch — pitch workflow
Portfolio — complete project archive	GitHub — source and experiments

LEARNING & TEACHING

- I created student-facing applied-computing material and practical programming exercises while teaching fundamentals through projects.
- I built The OG Roadmap · The Complete Book: a six-part self-directed inference-engineering curriculum with a 157-item implementation gauntlet covering mathematics, PyTorch, serving, C++ systems, CUDA/Triton, evaluation, and contribution.

TOOLS I USE

Systems	Python · Rust · Go · SQL	ML runtime	PyTorch · Transformers · CUDA measurement
Software	TypeScript · React · Next.js · Node.js · PostgreSQL · Redis · Docker		